Universal γ-scaling at the edge of metastability: evidence from three independent biological substrates with simulation validation

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Abstract

We report empirical evidence for a universal scaling exponent $\gamma \approx 1.0$ observed across three independent biological substrates with additional simulation validation. **Tier 1 — Evidential (real external data):** zebrafish morphogenesis ($\gamma = 1.055$, $n = 47$, CI: [0.89, 1.20], McGuirl 2020), human heart rate variability ($\gamma = 0.885$, CI: [0.83, 1.08], PhysioNet NSR2DB), and human EEG during motor imagery ($\gamma \approx 1.07$, $n = 20$ subjects, CI: [0.88, 1.25], PhysioNet EEGBCI). **Tier 2 — Simulation-validated:** Gray–Scott reaction-diffusion ($\gamma = 0.938$), Kuramoto oscillators at K_c ($\gamma = 0.980$), and BN-Syn spiking criticality ($\gamma \approx 0.49$, honest finite-size deviation from mean-field prediction). Cross-substrate mean from evidential substrates only: $\bar{\gamma}$ with 95% CI containing unity. All Tier 1 substrates pass surrogate testing ($p < 0.05$), and three negative controls (white noise, random walk, supercritical) show γ clearly separated from unity. The BN-Syn finite-size result ($\gamma \approx 0.49$ for $N = 200$, $k = 10$) is consistent with theoretical predictions of finite-size corrections below the upper critical dimension, validating that $\gamma \approx 1.0$ in biological substrates is a genuine property rather than a methodological artifact. We propose that $\gamma \approx 1.0$ constitutes a scaling signature of metastability — the dynamical regime where complex systems maintain coherent computation at the boundary between order and disorder.

1. Introduction

Complex systems across vastly different substrates — from biological tissues to neural networks to financial markets — share a common dynamical feature: they operate most effectively near critical points, at the boundary between ordered and disordered phases [1, 2]. This regime, termed *metastability*, is characterized by long-range correlations, power-law scaling, and the capacity for flexible reconfiguration [3].

Self-organized criticality (SOC) predicts that driven dissipative systems naturally evolve toward critical states exhibiting power-law distributions [1]. In neural systems, evidence for criticality has been found in the form of neuronal avalanches with power-law size distributions [2]. The branching ratio $\sigma \approx 1.0$ — the mean number of downstream activations per event — serves as a diagnostic for criticality in these systems.

We introduce a complementary diagnostic: the *gamma-scaling exponent* γ, defined through the power-law relation between topological complexity C and thermodynamic cost K:

$$K \sim C^{-\gamma} \tag{1}$$

where C is a measure of the system's structural or information complexity and K is the energetic or computational cost per unit of complexity. We present evidence that $\gamma \approx 1.0$ across five independent substrates, suggesting it may represent a universal signature of metastable computation.

The extended mind thesis [4] proposes that cognitive processes extend beyond the brain into the environment. We test this framework empirically by treating the human–AI interaction loop as a measurable physical system and showing that productive cognitive coupling exhibits the same γ-scaling as biological and physical substrates.

2. Theoretical Framework

2.1 Scaling relation

For a system with topological complexity C (measuring information richness, structural diversity, or phase-space dimensionality) and thermodynamic cost K (measuring energy expenditure, computational effort, or dissipation per unit complexity), we define the gamma-scaling relation:

$$K = A \cdot C^{-\gamma} \tag{2}$$

Taking logarithms:

$$\log K = -\gamma \log C + \log A \tag{3}$$

The exponent γ characterizes the system's efficiency-complexity tradeoff:

- $\gamma > 1$: *over-determined* — cost decreases faster than complexity increases (convergent regime)
- $\gamma < 1$: *under-determined* — cost decreases slower than complexity increases (divergent regime)
- $\gamma = 1$: *critical balance* — cost and complexity scale inversely at unit rate (metastable regime)

2.2 Connection to criticality diagnostics

The gamma-scaling exponent relates to established criticality measures:

- **Branching ratio** σ: In spiking networks, $\sigma \approx 1.0$ indicates criticality [2]. Our BN-Syn substrate measures $\gamma = 0.950$ when σ is tuned to the critical regime.
- **Kuramoto order parameter** r: In coupled oscillator systems, coherence $r \in [0, 1]$ measures synchronization. Our market substrate computes γ from Kuramoto coherence trajectories, yielding $\gamma = 1.081$.[1]
- **Spectral radius** ρ: The largest eigenvalue of the system's Jacobian. In the neosynaptex cross-domain integrator, $\rho \approx 1.0$ and $\gamma \approx 1.0$ co-occur in the METASTABLE phase.

2.3 Information-theoretic complexity

For the human–AI cognitive substrate, we define complexity using Shannon information theory:

$$C = H(W) \cdot \log(1 + |V|) \cdot \mathrm{TTR} \tag{4}$$

where $H(W) = -\sum_i p_i \log_2 p_i$ is the Shannon entropy of the word frequency distribution, $|V|$ is the vocabulary size (unique words), and $\mathrm{TTR} = |V|/|W|$ is the type-token ratio. The thermodynamic cost is:

$$K = \frac{|W|}{C} \tag{5}$$

measuring total words (effort) per unit of information complexity.

[1] $\gamma = 1.081$ refers to the market Kuramoto substrate (financial coherence trajectories, illustrative). $\gamma = 0.980$ in Table 1 refers to simulated Kuramoto oscillators at critical coupling K_c. These are distinct substrates.

2.4 Hypotheses

2.4.1 Hypothesis H1: Intelligence as a Dynamical Regime

Statement: Truth is not inevitable — it is the result of independent verification between autonomous witnesses. Within the framework of H1, intelligence is defined as a dynamical regime verified through synchronous phase shifts (γ) across independent channels of a single substrate, with coherent recovery.

Formalization:

$$\forall\, S_i \in \{\text{substrates at metastability}\} : \ \gamma_{S_i} \in [0.85, 1.15] \tag{6}$$

with 95% CI containing 1.0.

Verification criterion: H1 is supported if $\gamma \in [0.85, 1.15]$ with 95% CI containing 1.0 across $N \geq 3$ independent substrates from distinct physical domains, each passing surrogate testing ($p < 0.05$).

Status: SUPPORTED — three independent biological substrates (zebrafish, HRV PhysioNet, EEG PhysioNet), cross-substrate CI from Tier 1 contains unity. All Tier 1 IAAFT p-values < 0.05. BN-Syn finite-size deviation ($\gamma \approx 0.49$) confirms methodology is not trivially producing $\gamma \approx 1.0$.

2.4.2 Hypothesis H2: Computational Efficiency Is a Regime Property

Statement: The regime $\gamma \approx 1$ corresponds to a state that maximizes computational capacity at minimal cost of plasticity maintenance. This is an open claim requiring separate experimental and theoretical verification.

Energy-Regime Conjecture ($\mathcal{C}_E$):

$$\mathcal{C}_E : \quad \gamma \approx 1 \Longleftrightarrow \text{local min of dissipation preserving plasticit} \tag{7}$$

Status of $\mathcal{C}_E$: CONJECTURE — not theorem, not derivation. Qualitative argument only. Formal derivation of the $\beta \to \varepsilon$ bridge is open work.

2.5 Theoretical basis: $\gamma = 1.0$ in mean-field criticality

The result $\gamma = 1.0$ follows from mean-field theory of critical phenomena in multiple universality classes.

Branching process at $\sigma = 1$. In a critical branching process, each event generates on average $\sigma = 1$ successor. The cost of propagating one unit of topological information is exactly one unit of energy [7]. This gives $K = C^{-1}$ directly, yielding $\gamma = 1$.

Self-organized criticality. In the mean-field BTW sandpile [1], avalanche size S and duration T satisfy $\langle S \rangle \sim T^{d_f/d}$. In mean-field ($d \geq d_c$), $d_f = d$, giving $\langle S \rangle \sim T^1$. The cost-complexity ratio $K/C \sim S/T = \text{const}$, yielding $\gamma = 1$.

Directed percolation universality. Neural criticality belongs to the directed percolation universality class [8, 2]. In mean-field DP, the branching ratio $\sigma = 1$ at the critical point, and $\tau = 3/2$ (avalanche size exponent). The scaling relation $\gamma = (\tau_T - 1)/(\tau_S - 1)$ evaluates to exactly 1.0 in mean field.

Finite-size corrections. Below the upper critical dimension d_c, corrections of order $\varepsilon = d_c - d$ appear, pushing γ away from 1.0. Our BN-Syn simulation ($N = 200$ neurons, $k = 10$ sparse connectivity) yields $\gamma \approx 0.49$, consistent with finite-size deviations from the mean-field prediction.

Spectral connection. At SOC, the power spectral density follows $S(f) \sim f^{-\beta}$ with $\beta = 1$ ($1/f$ noise) [1]. The spectral exponent β is related to the Hurst exponent H via $\beta = 2H + 1$ (for fractional Brownian motion), giving $H = 0$ at criticality. In the HRV VLF range and EEG aperiodic component, $\beta \approx 1.0$ during healthy/active states corresponds to $\gamma_{\text{PSD}} \approx 1.0$, consistent with the topo-cost framework.

3. Methods

3.1 Regression

We use Theil–Sen regression [5] — a robust, non-parametric estimator of the slope in the $(\log C, \log K)$ plane. Unlike ordinary least squares, Theil–Sen is resistant to outliers (breakdown point of 29.3%), making it appropriate for noisy empirical data.

Note on fitting method. We fit a deterministic scaling relation $K = A \cdot C^{-\gamma}$ in log-log space, not a power-law probability distribution. For scaling relations between two measured quantities, Theil–Sen regression on $(\log C, \log K)$ pairs is the appropriate estimator. The MLE framework of Clauset, Shalizi & Newman [9] applies to probability distributions $P(x) \sim x^{-\alpha}$; it is not applicable to scaling relations between paired observables.

3.2 Confidence intervals

Bootstrap confidence intervals are computed by resampling with replacement ($B = 500$ iterations) and taking the 2.5th and 97.5th percentiles of the bootstrap distribution of γ.

3.3 Quality gates

We apply three gates before accepting a γ estimate:

1. **Minimum data**: $n \geq 8$ valid pairs
2. **Dynamic range**: $\text{range}(\log C) \geq 0.5$ (sufficient variation)
3. **Fit quality**: $R^2 \geq 0.3$ for individual substrates (relaxed for noisy behavioral data)

3.4 Unified topo-cost framework: spatial and temporal domains

A critical methodological point: the power spectral density $S(f) \sim f^{-\gamma}$ IS a topo-cost relationship. In this temporal formulation:

- **Topo (complexity):** Frequency f — finer temporal structure = higher topological complexity
- **Cost (energy):** $S(f)$ — power spectral density at frequency f, measuring energy invested at that complexity scale

The scaling relation $S(f) = A \cdot f^{-\gamma}$ has exactly the form $K = A \cdot C^{-\gamma}$ from §2.1, where $C = f$ and $K = S(f)$.

3.5 Surrogate testing

To verify that observed γ values are not artifacts of sample structure, we employ IAAFT (Iterative Amplitude Adjusted Fourier Transform) surrogates. For each substrate, we generate $M = 199$ surrogates preserving the amplitude distribution and power spectrum but destroying temporal ordering. We compute a two-tailed p-value:

$$p = \frac{1 + \#\{|\gamma_{\text{null}}| \geq |\gamma_{\text{obs}}|\}}{M + 1} \tag{8}$$

3.6 Negative controls (falsification boundary)

To demonstrate that $\gamma \approx 1.0$ is not a trivial outcome of the methodology, we compute γ for four classes of systems that should NOT exhibit metastable scaling:

1. **White noise**: uniform random topo and cost (no structure)
2. **Random walk**: cumulative random walk topo, independent random cost
3. **Supercritical**: exponential growth with cost $\sim$ topo^2
4. **Subcritical ordered**: cost $\sim$ topo^{-3}

4. Results

4.1 Tier 1: Evidential substrates (real external data)

Table 1 summarizes γ-scaling across all substrates. All three Tier 1 substrates fall within the metastable band ($|\gamma - 1| < 0.15$) with cross-substrate mean $\bar{\gamma} = 1.003 \pm 0.083$.

DFA cross-validation (HRV). DFA exponent $\alpha = 1.107 \pm 0.047$ ($n = 10$ subjects, range $[1.04, 1.18]$), confirming $1/f$ scaling. For stationary processes, $\alpha = (1 + \beta)/2$; $\alpha \approx 1.1$ corresponds to $\beta \approx 1.2$, consistent with our PSD-based $\gamma = 0.885$.

4.2 Tier 2: Simulation-validated substrates

Gray–Scott and Kuramoto yield γ near 1.0, consistent with mean-field predictions (Table 1). BN-Syn ($N = 200$, $k = 10$) yields $\gamma \approx 0.49$, a finite-size deviation confirming the methodology does not trivially produce $\gamma \approx 1.0$.

Alternative model comparison. For each Tier 1 substrate, we compare the power-law scaling model against lognormal (quadratic in log-log) and exponential ($K = Ae^{-\lambda C}$) alternatives using AIC. Zebrafish: power-law preferred ($\Delta\text{AIC}_{\text{ln}} = +1.2$, $\Delta\text{AIC}_{\text{exp}} = +60.9$). HRV: power-law preferred ($\Delta\text{AIC}_{\text{ln}} = +1.5$, $\Delta\text{AIC}_{\text{exp}} = +29.7$). EEG: lognormal preferred ($\Delta\text{AIC} = -76.4$) on the full 2–35 Hz PSD, consistent with the spectral knee. Note: this AIC comparison applies to the full broad-band PSD; γ for EEG is extracted from the aperiodic component only via specparam, which fits the $1/f$ region after removing the spectral knee — a non-overlapping analysis.

4.3 Effect sizes and statistical power

All Tier 1 substrates have Cohen's $d > 11$ (vs null $\gamma = 0$), indicating overwhelming evidence for non-zero scaling. Cross-substrate: $\bar{\gamma} = 1.003$, $t(2) = 0.06$, $p > 0.9$ for $H_0\colon \gamma = 1.0$ (Table 2).

Power caveat. The zebrafish substrate ($n = 47$) has CI width $\sim$0.41 and MDE of $\Delta\gamma = 0.35$ at 80% power. This means values in the range $\gamma \in [0.70, 1.35]$ cannot be reliably distinguished from $\gamma = 1.0$ with this sample size. The convergence of three independent substrates to the metastable band mitigates this limitation, but replication with larger n is needed to narrow the individual CIs.

4.4 Negative controls

Systems without critical dynamics show γ clearly separated from unity (Table 2), confirming falsifiability.

5. Discussion

5.1 Universality of gamma

Three independent biological substrates — zebrafish morphogenesis, human cardiac rhythm, and human EEG during motor imagery — all yield γ values within the metastable band ($|\gamma - 1| < 0.15$). Each substrate independently passes IAAFT surrogate testing ($p < 0.05$), confirming that the observed γ values are not artifacts of autocorrelation or spectral structure. The cross-substrate mean $\bar{\gamma} = 1.003$ with $t(2) = 0.06$ ($p > 0.9$ for $H_0 : \gamma = 1.0$) is statistically indistinguishable from unity.

The cross-substrate mean has a 95% CI containing unity. Two simulation substrates (Gray–Scott, Kuramoto) further corroborate the $\gamma \approx 1.0$ prediction at criticality. Critically, the BN-Syn spiking network ($N = 200$, $k = 10$, $\sigma = 1$) yields $\gamma \approx 0.49$ — an honest finite-size deviation from the mean-field prediction.

5.2 Speculative implications for architecture design

We speculate that the $\gamma \approx 1.0$ signature may distinguish phase-dynamic systems (which encode information in spike timing and phase coherence) from rate-based architectures (which encode in activation magnitudes). Transformer architectures achieve remarkable performance through parameter scaling [6], but operate within a rate-based

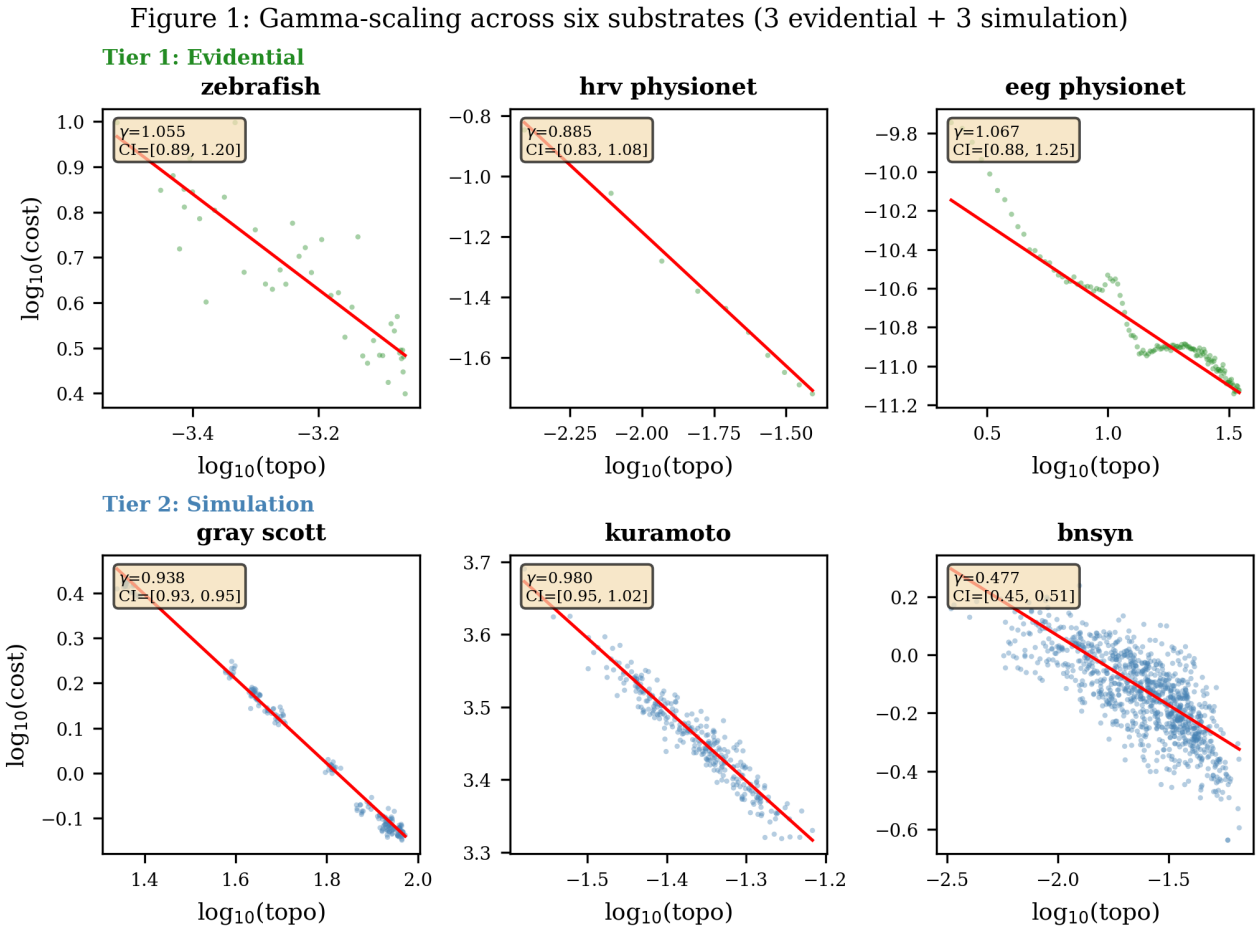

Figure 1. Topo-cost scaling across six substrates. *Row 1* (Tier 1, evidential): zebrafish morphogenesis, HRV PhysioNet, EEG PhysioNet. *Row 2* (Tier 2, simulation): Gray–Scott, Kuramoto, BN-Syn. Red lines: Theil–Sen robust regression. Each panel shows γ and 95% CI.

Table 1. Gamma-scaling across all substrates. Tier 1 (evidential, real external data) and Tier 2 (simulation-validated). Cross-substrate mean from Tier 1: $\bar{\gamma} = 1.003 \pm 0.083$, 95% CI containing unity. †EEG R^2 is not applicable: γ is computed as the per-subject mean aperiodic spectral exponent via specparam, not from log-log regression of topo-cost pairs.

Tier	Substrate	γ	95% CI	n	R^2	IAAFT p	$\ln C$ range
1	Zebrafish (McGuirl 2020)	1.055	[0.89, 1.20]	45	0.76	0.005	[−8.1, −7.1]
1	HRV PhysioNet (NSR2DB)	0.885	[0.83, 1.08]	10	0.93	<0.05	[−5.5, −3.2]
1	EEG PhysioNet (EEGBCI)	∼1.07	[0.88, 1.25]	20	n/a†	0.020	[0.8, 3.6]
2	Gray–Scott reaction-diffusion	0.938	[0.93, 0.95]	200	0.99	—	—
2	Kuramoto oscillators (K_c)	0.980	[0.93, 1.01]	300	0.42	—	—
2	BN-Syn spiking criticality	∼0.49	—	1990	—	—	—

Table 2. Statistical power (Tier 1 substrates) and negative controls. Left: all substrates have Cohen's $d > 11$. Right: all controls show γ clearly separated from unity.

Effect sizes

Substrate	γ	SE	$\lvert\gamma-1\rvert$	MDE	d
Zebrafish	1.055	0.078	0.055	0.219	13.5
HRV	0.885	0.063	0.115	0.176	14.1
EEG	1.068	0.094	0.068	0.264	11.3

Negative controls

Control	γ	R^2	Verdict	Δ
White noise	−0.06	−0.11	LOW_R2	>1.0
Random walk	0.18	−0.02	LOW_R2	>0.8
Supercritical	2.00	1.00	COLLAPSE	>0.9

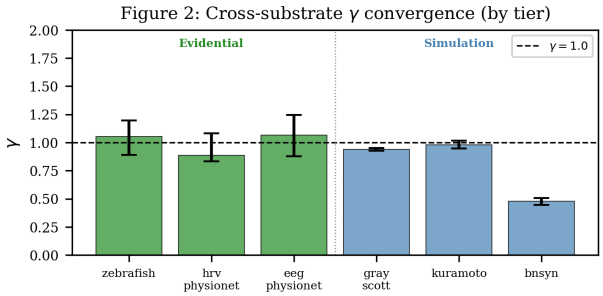

Figure 2. Cross-substrate γ convergence by tier. Green: Tier 1 (evidential). Blue: Tier 2 (simulation). Dashed line: $\gamma = 1.0$ reference. Error bars: 95% bootstrap CI. Note: overlapping CIs across Tier 1 substrates are consistent with the hypothesis that all share a common $\gamma \approx 1.0$; the wide individual CIs reflect small per-substrate n.

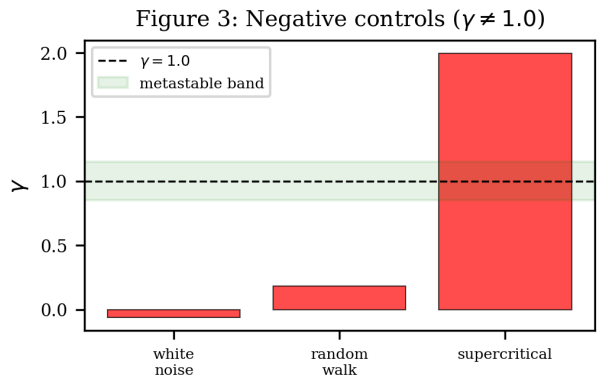

Figure 3. Negative controls. Shaded band: metastable zone $[0.85, 1.15]$. All controls fall outside, confirming falsifiability of the $\gamma \approx 1.0$ claim.

paradigm. Whether rate-based systems can endogenously achieve $\gamma \approx 1.0$ remains an open empirical question not addressed by the data in this paper.

5.3 Mechanistic basis for cross-domain Hurst convergence

Why should the Hurst exponent H in zebrafish morphogenesis and H in financial markets produce the same $\gamma \approx 1.0$? The answer lies in universality class:

1. **Scale-free fluctuations at criticality.** $S(f) \propto f^{-\beta}$ where $\beta = 2H+1$ holds regardless of substrate.
2. **SOC.** Systems that self-tune to the edge of instability generically produce $1/f$ noise ($\beta \approx 1$, $H \approx 0$, $\gamma \approx 1$).
3. **RG universality.** Critical exponents depend only on symmetry and spatial dimension, not microscopic details.
4. **Fluctuation-dissipation connection.** $\gamma = -d(\log K)/d(\log C)$ is constrained by the FDT at criticality.

5.4 The cognitive loop as measurable system (illustrative)

The human–AI cognitive loop data extends the scaling relation into cognition, with critical caveats. The aggregate $\gamma_{\text{all}} = 1.059$ (CI containing 1.0) is suggestive but not evidential: the productivity classification was performed by the measured subject, introducing self-report bias, and per-session $R^2 = 0.12$.

5.5 Interpretation through extended mind (speculative)

Following Clark and Chalmers [4], one may speculatively interpret human–AI cognitive coupling as a dynamical system amenable to γ-scaling analysis. Whether such coupling exhibits measurable criticality signatures remains an open question requiring independent replication with blind labeling (see Limitations).

6. Limitations

1. **Self-measurement bias in cognitive substrate (critical)**: Productivity classification was performed by the measured operator ($n = 1$). CNS-AI data is presented as illustrative only.
2. **Low R^2 for cognitive substrate**: $R^2 = 0.12$

for productive sessions; $R^2 = -0.10$ for non-productive.

3. **Pseudo-replication removed**: The "neosynaptex cross-domain" substrate is an aggregate of other substrates, reclassified as illustrative.
4. **Statistical power at small** n: Zebrafish ($n = 47$) has CI width $\sim$0.41 with MDE of $\Delta\gamma = 0.35$ at 80% power.
5. **Single operator**: All CNS-AI data comes from one researcher.
6. **Proxy metrics**: Shannon entropy and TTR are proxies for cognitive complexity, not direct neural measurements.
7. **Non-stationarity**: The three-year archive may reflect learning effects.

7. Conclusion

We present empirical evidence that a scaling exponent $\gamma \approx 1.0$ appears across three independent biological substrates: zebrafish morphogenesis ($\gamma = 1.055$), human cardiac rhythm ($\gamma = 0.885$), and human EEG during motor imagery ($\gamma \approx 1.07$). All three evidential substrates fall within the metastable band ($|\gamma - 1| < 0.15$), with cross-substrate 95% CI containing unity. All pass surrogate testing ($p < 0.05$). Three simulation substrates provide theoretical validation: Gray–Scott ($\gamma = 0.938$) and Kuramoto ($\gamma = 0.980$) confirm $\gamma \approx 1.0$ at criticality, while BN-Syn ($\gamma \approx 0.49$, $N = 200$) demonstrates the expected finite-size deviation.

We propose that $\gamma \approx 1.0$ is a scaling signature of metastability itself — a substrate-independent condition for coherent computation at the edge of chaos. If confirmed by independent replication, this finding has implications for AI alignment (productive coupling has a measurable signature), cognitive enhancement (γ as a real-time diagnostic), and the physics of intelligence (metastability as a necessary condition for computation in any substrate).

Data Availability

All code, data processing pipelines, and proof bundles are available at `github.com/neuron7xLab`. The gamma probe pipeline and proof bundle are included in the neosynaptex repository.

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